

Living Data Systems: Co-Designing Community-Based Methods And Local Technologies For Inclusive Socioeconomic Alternatives

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ABSTRACT

This research project aims to collaborate with underserved communities leading social justice efforts through alternative economic practices, in the co-creation of a data model based on sets that members can collect to represent meaningful, contextual and social information. Data, understood from a cooperative-value and qualitative perspective, could be interesting for the common use of the group to interpret and exchange through non-monetary transactions, in contrast to the predominantly capital value mainstream networks. This system could facilitate group conversations on collective appropriation, ownership and adaptation processes through the co-design of local, situated technologies and alternative outputs of information. Furthermore, it could be significant for existing practices and goals around socioeconomic change, autonomy or activism.

It is necessary to research the implementation of novel data collecting methods and data-driven systems focused on non-traditional, qualitative and plural representations of knowledge. This could be instrumental for understanding how to visualise and integrate omitted, hidden or underrepresented experiences, how to include marginalised individuals with different backgrounds into diverse decision-making processes and how to co-design new technologies that account for shared values. On a deeper level, it could help uncover new data epistemologies and definitions emerging from the construction of new collective knowledge platforms, understood as living systems.

CCS CONCEPTS

• Human-centered computing; • Interaction Design; • Participatory Design;

KEYWORDS

Design Justice, Design Research, Data Commons, Human Data Interactions, Community Economics

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1 BACKGROUND AND MOTIVATION

Data has facilitated the development of platforms that people interact with daily. While such advances are favoured for efficiency and convenience, they often also sustain ineffective and problematic processes for accessing resources, perpetuating stereotypes, promoting exclusion and narrowing decisions and operations for particular groups of people. Mainstream technologies, in many cases, are designed and developed from a non-neutral and non-plural perspective, replicating patterns and models of inequality that exclude individuals at the margins [11], generally or historically affected by power structures.

Alternative social systems and knowledges, as well as their practitioners, might be underrepresented or completely excluded within the strict and hegemonic characteristics that have been given to data, where local imaginaries might not be reflected and voices can be indirectly silenced by the hierarchical and standardised manner of developing mass technologies. Under such circumstances, information technologies and design should be collaborators for responsible innovation and more accessible and inclusive tools [3] that promote new forms of meaningful, fair and ethical connections and data interactions that represent diverse realities.

Based on this premise, data itself should be collected, produced, shared and understood from the same approach: a plural, diverse and multi-sided value perspective that integrates the properties it has outside the capital-aimed ones and that does not ignore or exclude the different lived experiences and backgrounds of people [39]. Therefore, it is indispensable to co-create novel models, technologies and practical methods that allow information to flow in equal directions, through transparent interactions.

My research aims to acknowledge and recognise the significance of existing practices or activities currently under-valued, and which are relevant for people in relation to their communal and daily dynamics. The translation of these actions into a co-created model could help obtain insights on how alternative approaches to data can make it more locally empowering for collective interests. This could enable ethical and responsible data-driven practices that support fairer values, similar to what existing alternative models as community economics [19], cooperative economics [21], degrowth [27] and commoning [16, 20] propose.

The relevance for my project to collaborate with an organisation which is currently performing practices related to alternative economics, relies on the practical incidence of co-researching new ideations that can be applied to an existing living social system that is by itself shaping its own socio-political reality. This provides a background for experimenting with situated technologies [14] as tools to incite a needed “transformational shift from quantitative to qualitative development” [44: p. 187]. Design practice, therefore,

should “contribute to broadening, and drawing on the rich spectrum of experiences that should be considered viable alternatives to what exists” [14: p. 92] from a lens that moves away from the Westernised production models. This could lead to develop new epistemologies around the concept of data, adding to the conversations for intersectionality [8] and decolonial data [5, 34] to shape new qualitative technologies.

2 RELATED WORK

The aspiration of engaging in a co-creative development with individuals at the socioeconomic margins, relies on their current lack of representation in digital-driven efforts and mainstream artifacts. They might suffer oppressions from a sociotechnical structure that is not neutral [3, 4], where particular abilities, knowledges and other factors might not be integrated. I have identified three pillars as key-stones for my project, driven by embracing plural knowledges and contextual values, thoughtfully interpreted into the topics of social justice (focused on design justice [3] and HCI justice), alternative economics and alternative data.

The literature review has been particularly focused on research projects intersecting the aforementioned disciplines understood from HCI frameworks. Co-creative projects have provided a grounding frame to create awareness around the politics and logics that rule the design of algorithms and digital services, showing a series of initiatives around fairness and justice in technology [9]. In a similar spirit, DIY networking efforts have enabled the collective awareness of communities in physical proximities, allowing for particular contents to be shared, promoting different socio-economic ideations [45]. Collaborative work with communities related to HCI and justice can also be seen in the contexts of both the Global North and Global South with proposals that are highly practice-based and which allow for reflection on the outcomes that experimentations at this scale have provoked [26, 35, 37, 43, 48], while others have addressed HCI, solidary movements and economy [47].

Similar initiatives, focused on the use of DIY sensors for datafying contextual social issues, have denoted community engagement and technological appropriation [6, 28, 49, 50]. Despite their relevance in creating awareness and a more democratic access to information, such tools have limitations in capturing and communicating data beyond its quantitative aspects, excluding layers of particular meanings and more qualitative sources. These topics have been analysed in efforts as mapping initiatives [7], but the information has been seldom interpreted as part of a data model or as a system of shared values.

The discussion on how data models could benefit the design justice and alternative economics frameworks is an underexplored area. The current practices and examples related to technology have not been around possibilities for qualitative values, whilst there exists an identified gap in the economic angle of Sustainable HCI [12, 41]. This provides an interesting opportunity to analyse how existing communal practices, new power structures and alternative representations in the pursue of socio-economic change could be analysed with the introduction of experimental platforms. Their use could help visualise justice by enabling a group of individuals in creating their own counter-data and their own descriptions of the concept.

From that perspective, the current presence of alternative economics symbolises that new political and epistemological approaches are possible and that they can be achieved and co-created. There exist a growing number of efforts in the HCI community, as platform cooperativism [33], visions for post-capitalism HCI [17] and the exploration of alternative futures [30]. Outside of the discipline, there have been several calls of action for postcapitalist narratives to emerge and thrive [13, 38]. Authors like J.K. Gibson-Graham [20], have already advocated for alternative practices as strategies to transform the capital-centric panorama and move beyond what quantitative and production-driven methods can account for. Hence, there is need for new metrics to include the plurality of activities and values involved in such living processes. This is also evoked by Escobar [14], who addresses the urgency of proposing design-led transitions toward more sustainable futures based on the communal and on privileging the diversity of knowledges outside of the Westernised ones. Discarding the mainstream, linear ideas of progress [14] in favour of more plural transitions [24, 32], could lead to healthier interrelations on the different nodes of economic, social and political complex systems.

The imaginaries and idiosyncrasies of decolonial and post-extractive systems should be considered pillars for interdisciplinary and co-creative HCI proposals and for constructing new frameworks driven by localised socio-economic justice. They could reorient the core concepts and objectives of technological dynamics that follow the same capitalistic quantitative-driven models of production and sensemaking. Efforts exist in areas as postcolonial computing [23] but there are opportunities to further explore the integration of technologies and HCI conceptualised from alternative epistemologies. This presents an opening to learn from practices based on justice, cooperation and non-competitive frameworks and to explore how particular aspects and sources around data could ambivalently affect existing systems or could support people in shaping relations around them.

3 RESEARCH GOALS

- I will analyse how a community-led and community owned information system grounded on social justice initiatives and alternative socio-economic practices, could contribute to sharing local knowledges, to tangibilise different social practices and to become an instrument for empowerment, prioritising underrepresented groups and imaginaries.
- I will study how current socio-economic and cultural practices attached to alternative systems - regardless of their production value - are translated and represented through new definitions of socially-driven data models and epistemologies around the meanings of data itself.
- I will explore and reflect on the pertinence of alternative representations of data as methods to interpret and make sense of certain aspects of complex social systems and existing community-led practices and intangible values through qualitative sensing explorations, in order to co-design impactful actions, analyse its effects on existing systems or create new relations within the members of a specific group.

In connection to these aims and objectives, my research questions are:

- How can collaborative, fair and inclusive systems be developed more meaningfully to support the exchange of non-monetary values and further cooperation among members of a group?
- How can novel participatory, speculative design methods facilitate local communities co-creating, appropriating and interpreting alternative data models?
- How can collective data engagement practices help uncover activities, transactions and new exchange dynamics to support ethical socio-economic practices?
- How can data-driven prototypes support the meaningful translation of intangible meanings, qualitative factors and hidden practices into shared values?
- How can plural narratives and representations help create new data epistemologies around the interpretations of new forms of collective knowledge?

4 RESEARCH APPROACH, METHODS AND RATIONALE

The practice-based research will include a series of workshops, digital/physical experimentation, design probes and other speculative methods for the collaborative development of situated technologies. As opposed to measuring quantitative assets, these could reflect on the more intangible, rich and omitted factors that might not be currently represented elsewhere, but that are still invaluable parts of a community [2, 22, 29, 46].

In the context of the project, co-design will be approached through a series of collective ideations applied throughout the design process [39]. This will support the integration of traditional formations and collections as core values to be translated and embedded in the design of new systems. It will shift the hierarchic paradigm of digital access from the standard and centralised to the collective and participatory. Therefore, it will allow every actor involved in a local effort to see an extract of their own experience represented and to contribute, share and understand their own sources of data as part of a greater collaborative research and generative effort.

Whilst I envision a model for diagnosing intersectionality and justice, there are negative effects that the presence of this system, seen as an accessible resource for a particular group, could stimulate. I recognise the plausible risk of misuse practices that could latently lead to reductionist, tokenising and extractive actions that further amplify the inequalities of the community. Thus, it is fundamental to ideate strategies that allow for equal practices and accessibility within the members, and for constant dialogues among the group, to debate what intrinsically needs or does not need to be represented as/with/through data.

Furthermore, although privilege recognition and respect for traditional knowledges are concepts well integrated into the social justice discourse, it is important to acknowledge that the conventional definitions of economic development, social equity and environmental improvement are still grounded on hegemonic Westernised perceptions, and hence, taking a decolonial and anti-solutionist positionality is paramount.

I introduce the following approaches as co-creative and co-research methods to be shared with and repurposed by the community throughout the development of the project:

4.1 Qualitative sensing explorations

I suggest that new collaborative methods could be used to interpret particular intangible attributes [7] linked to social dynamics, from a shared perspective that contemplates the richness of non-quantitative factors [31]. Visualising non-measurable aspects related to hidden labour, care practices, non-monetary transactions and informal networks, for example, could lead to plural co-research practices with the community. Integrating speculative prototypes, here described as *qualitative sensing explorations*, could be helpful for experimenting with co-creative processes as means for individuals to propose layers of alternative data, collected from more sensory, value-driven and experiential lenses, and which could be relevant to them in their immediate context [10, 15].

Developing novel explorations (analogue or technological) based on communal needs, behaviours, curiosity and social factors, could provide participants with a more familiar interpretation of non-extractive data [5]. Recognising themselves as the owners and creators of such, could support in creating more contextually rich translations, feeding the prototypical model and the local community of data with a self-owned collection that could help envision ideas and practices. The resulting prototypes or artifacts could be useful for generating social, cultural and technical dialogues [1, 25, 40]. Once an unnoticed activity becomes data, it could completely change the way it is thought about and reflected upon. This could also provoke conversations related to what would happen if such data was effectively visualised and accessible to the rest of the group and what contextual consequences could the presence of this system stimulate.

4.2 A living data system

The conceptualisation of the proposed data-driven model as a living data system emerges from the living-systems theory, “a body of knowledge that explains the dynamics of self-organization, emergence, and resilience taking place within natural and social systems” [14: p.177]. A responsible and value-driven process at this scale could translate into a ‘living’ data system: a responsible and responsive tool that is released, unfinished, into its context. Hopefully, it will become part of the community, being led, repurposed and controlled cooperatively by its members to detect locally relevant insights for achieving common goals. Once the community-owned data is shared, the group could propose and lead to (a) co-create their own narratives and more equitable models of distribution for plural and diverse practices, (b) find new patterns for activism and incidence and (c) uncover positive and negative behaviours in their current power relations.

To contribute to these purposes, I suggest that the generative methods and explorations must enable for ambiguity [18] and for different interpretations to emerge, in contrast to a single or fully finalised outcome. Such attributes, translated into the forms of unfinished prototypes [36, 42], could be used to collect and take actions from a local pool of information, created, governed and managed by the community itself. Eventually, the use of these new methods could lead the group to an iterative process for their adoption and repurposing to facilitate innovative ideations. Based on the idea that particular tangible/intangible aspects around data can affect the system or can shape it, there exists a possibility for

transformative, and value-driven relations to emerge. Hence, the proposed model could become a situated technology “that, rather than being decontextualized and value neutral, is embodied, place based, convivial, and conducive to care.” [14: p.58]

5 RESULTS AND CONTRIBUTIONS TO DATE

In the nine months since starting my PhD, I have worked on the contextual and literature review for the research and I have explored different sectors relevant for my direction. I have arranged two meetings with representatives from an organisation in Mexico focused on developing participatory projects with marginalised indigenous populations for the creation of collaborative approaches, designed with the particular conceptions, values, and needs of the communities in mind. Coming from the Global South myself, it is pertinent for me to understand the areas in which my project could be developed and the incidence and impact it could have. Simultaneously, the plural sociocultural context of Mexico presents a rich background for alternative conceptions and decolonial knowledges, which could be respectfully addressed and reinterpreted as foundations for the co-creation of new data definitions and technologies based on such representations.

Parallel to this, I have visited communities and underserved groups in Scotland, specifically at the deprived neighbourhoods of Wester Hailes in Edinburgh, and Govanhill in Glasgow. I made contact and met with an activist and community leader in each location which proved to be valuable for starting conversations to understand the current context and needs that could lead to meaningful research collaborations for my PhD.

Approaching community/activist practices in the context of the UK and Mexico has helped me acknowledge the systemic, political and historic differences that configure complex sociotechnical issues in both locations and, therefore, the scopes and challenges each area could present. Considering trans-local connections between communities and organisations may have potential to be explored further throughout my PhD, opening the possibilities for implementing the insights and findings from the proposed methods around new data epistemologies, qualitative-driven models or even decolonial technologic concepts into the practices and ongoing efforts of communities in the context of the Global North.

6 NEXT STEPS, CHALLENGES AND LONG-TERM GOALS

My next step consists of defining and developing the methods and co-creative approaches that will be used during fieldwork research. By the time when the consortium takes place, I will already have established a partnership with the final community to collaborate with, which will consequentially delineate certain tactics based on the specific context, needs, traditional knowledges, technologies and ongoing efforts that are pertinent for the group. Participating in the doctoral consortium would be beneficial for discussing and receiving advice on relevant methods and technological approaches for co-creation for representativity, plurality and alternative knowledges. In addition, I would find it valuable to discuss the technical and data-driven side of the project (including prototypes and metrics), in order to gain deeper understanding of examples or current developments in the HCI community relating to my above

outlined field of research. Coming from a service design industry background, this will ideally, help me push the practice forward, as these topics have proven to be relevant in previous consortiums and current HCI discourse.

By the end of my PhD, I envision to have co-designed an alternative data-driven model with a community and to have evaluated the use of the proposed methods to conceptualise new situated technologies as living data systems. I consider that the incorporation of civic, economic and cultural aspects that configure the lived experiences of underrepresented sectors is not only a need to be fulfilled to achieve some form of justice and recognition in the present time. Therefore, as a long-term goal, I intend to continue co-researching more inclusive, plural, community-led design methods and technologies that can challenge current mainstream structures and that can enable dialogues to propose more diverse and sustainable futures.

7 RESEARCH SITUATION AND DISSERTATION STATUS

I am a PhD student in the Institute for Design Informatics, at the University of Edinburgh, advised by Dr Bettina Nissen. I am currently in my first year and my completion date is at the end of 2024; the length of the programme for full-time students in the UK is three years. I will conduct the fieldwork research of my project during the second and third trimesters of 2023. I have not attended another doctoral consortium.

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